Estimating the Economic Cost of One of the World's Major Insect Pests, *Plutella xylostella* (Lepidoptera: Plutellidae): Just How Long is a Piece of String?

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Forum

Estimating the Economic Cost of One of the World's Major Insect Pests, *Plutella xylostella* (Lepidoptera: Plutellidae): Just How Long Is a Piece of String?

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ABSTRACT Since 1993, the annual worldwide cost of diamondback moth, *Plutella xylostella* (L.) (Lepidoptera: Plutellidae), control has been routinely quoted to be US$1 billion. This estimate requires updating and incorporation of yield losses to reflect current total costs of the pest to the world economy. We present an analysis that estimates what the present costs are likely to be based on a set of necessary, but reasoned, assumptions. We use an existing climate driven model for diamondback moth distribution and abundance, the Food and Agriculture Organization country *Brassica* crop production data and various management scenarios to bracket the cost estimates. The "length of the string" is somewhere between US$1.3 billion and US$2.3 billion based on management costs. However, if residual pest damage is included then the cost estimates will be even higher; a conservative estimate of 5% diamondback moth–induced yield loss to all crops adds another US$2.7 billion to the total costs associated with the pest. A conservative estimate of total costs associated with diamondback moth management is thus US$4 billion–US$5 billion. The lower bound represents rational decision making by pest managers based on diamondback moth abundance driven by climate only. The upper estimate is due to the more normal practice of weekly insecticide application to vegetable crops and the assumption that canola (*Brassica napus* L.) is treated with insecticide at least once during the crop cycle. Readers can decide for themselves what the real cost is likely to be because we provide country data for further interpretation. Our analysis suggests that greater efforts at implementation of even basic integrated pest management would reduce insecticide inputs considerably, reducing negative environmental impacts and saving many hundreds of millions of dollars annually.

KEY WORDS bioclimatic modeling, cost of management, pest control, diamondback moth, integrated pest management

Entomologists are notorious for conjuring monetary values to ascribe to pest problems when emphasizing the importance of a given pest in a scientific paper or justifying grant applications to sanction research to address the issue. The diamondback moth, *Plutella xylostella* (L.) (Lepidoptera: Plutellidae) is, at first glance, different. Ever since Talekar and Shelton (1993) published their landmark review, the annual cost of diamondback moth control to the world economy has been cited as US$1 billion by pretty much everyone writing a paper on diamondback moth, and no doubt when applying for grant funds. Interestingly the figure has not changed since 1993, even with inflation and increased production of *Brassica* hosts, particularly canola (*Brassica napus* L.). Perhaps pest management has become more efficient after all or maybe the original estimate was hyperbole? The figure itself was not calculated by Talekar and Shelton (1993) but was taken from a short foreword to the Proceedings of the Second International Workshop on Diamondback Moth and other Crucifer Pests (Talekar 1992) in which Javier (1992) suggested that "control could cost approximately U.S. $1 billion annually," but provided no justification for the estimate.

Here, we present a detailed basis for estimating the worldwide control costs of diamondback moth. Our aim is to outline a transparent methodology for arriving at a justifiable value that can be readily updated over time and that could be applied to other crop–pest systems. The need for such a monetary value can be justified on a number of dry economic rationalist grounds quite apart from its necessity for grant applications and the introductions to scientific manuscripts. Ultimately, applied research needs to redress major constraints to food security and so, of the myriad

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of competing interests, how do we objectively compare which pests are major constraints and justify research effort and allied expenditure? So that valid comparisons can be made, the inevitable assumptions must be justified and the methodology that is applied needs to be transparent.

To estimate the total cost of diamondback moth to *Brassica* crop production and management, we require some knowledge of how much is spent per ha of diamondback moth host plant (=*Brassica* crop) production on diamondback moth management and how much production is lost despite this management. These costs will depend in part on where diamondback moth is found (abundance varies geographically), how abundance is distributed relative to agricultural host plants and how much economic damage diamondback moth feeding inflicts on crops. None of these questions is easy to answer. Despite being such a major pest of crucifers (since 1985 six international workshops or conferences have been devoted to the pest (Talekar and Griggs 1986, Talekar 1992, Sivapragasam et al. 1997, Endersby and Ridland 2004, Shelton et al. 2008, Srinivasan et al. 2011) and being, perhaps, the most widely distributed lepidopteran species, our understanding of the geographic distribution and abundance of diamondback moth is limited to course scale and probably incorrect maps. The original Commonwealth Institute of Entomology map for diamondback moth (Commonwealth Institute of Entomology 1967) is an amalgam of incomplete distribution records and imagination (see Zalucki and Furlong 2008), whereas more recent CAB maps just show a dot in a whole country if the species is recorded (CAB 2012)! First, we present a way to estimate, albeit crudely, the distribution and abundance of diamondback moth globally. Next, we estimate a relationship between the suitability of a location for diamondback moth and the attendant pest management costs and, assuming the world is rational, insecticides are only applied when pest levels warrant control. We then use the production data for each country to bracket the likely costs given certain assumptions. The brackets will necessarily be wide and the "piece of string" is of variable length between these brackets!

Methods

Estimating the Distribution Of Diamondback Moth Abundance. Zalucki and Furlong (2008, 2011) developed a CLIMEX model (Sutherst and Maywald 2004) for diamondback moth that can be used to predict its geographic distribution and relative abundance, its seasonal phenology across its distribution range, and even long-term variations in abundance; see Zalucki and Furlong (2005) for details of the approach. The rationale under pinning CLIMEX has been described many times (Yonow and Sutherst 1998, Zalucki and Furlong 2008, Li et al. 2012), but we outline the approach briefly for readers not yet familiar with the methodology.

CLIMEX calculates a growth index (GI), analogous to population growth rate, that describes the potential of the population to increase at a location. The growth index, calculated weekly, is a product of temperature (TI) and moisture (MI) indices. For calculation of both TI and MI, there is a range of conditions of temperature and moisture over which growth is maximal. Either side of the optimum range growth rate decreases. Above some upper threshold and below a lower threshold growth ceases. These thresholds define the biological parameters for a species and need to be estimated (see, e.g., Zalucki and van Klinken 2006).

A measure of the relative suitability of a location for a species to persist is summarized in a single annual eco-climatic index (EI), scaled to 100: $EI = 100 \sum GI/52 \times SI \times SX$, where 52 is the number of weeks in a year; SI are four stress indices, describing the species response to the extremes of cold (CS, cold stress), heat (HS), dry (DS), and wet (WS) conditions; and SX, if needed, are interactions between extreme conditions; cold-dry (CDS), cold-wet (CWS), hot-dry (HDS), and hot-wet (HWS) stresses. The stress indices (SI and SX) are accumulated at a specified rate whenever conditions exceed a specified threshold.

Indices are calculated weekly for each location using standard meteorological data; a location's long-term monthly average maximum and minimum temperatures, rainfall, and humidity. Areas with positive values for EI are suitable for species persistence; the larger the value of EI, the more suitable the location. Generally, an EI of <15–20 would be considered marginal for a species' long-term persistence at a location. The values for parameters that describe the TI and MI components of growth, and the various stress indices (SI) and their interactions (SX), can be estimated from laboratory or field studies (Zalucki and van Klinken 2006).

Unknown or poorly measured values are estimated by an iterative procedure that involves comparing the predicted geographic distribution with the actual distribution and adjusting parameter values. Comparing the predicted and observed species distribution in an area (e.g., a continent) not used for the procedure can test the parameter tuning; see Maywald and Sutherst (1991) for details. This is one independent test of the model and can be contrasted with the more commonly used procedure of "randomly" subsampling from a current geographic distribution to both fit and test various empirical GIS models (e.g., as is done with MAXENT or similar), but of course comparisons across continents is only applicable if a species is widespread.

The CLIMEX model for diamondback moth in Zalucki and Furlong (2008) was initially developed using the anecdotal distribution data in Commonwealth Institute of Entomology maps (Commonwealth Institute of Entomology 1967), the seasonal prevalence of diamondback moth in the Cameron Highlands of Malaysia and published values for various parameters (Zalucki and Furlong 2008, 2011). There is no distinction between the growth and development responses of temperate and tropical diamondback moth populations to different temperatures (Shirai 2000), and it

Table 1. Field trials used to determine the no. of insecticide applications used in IPM or FP control strategies

Location	Crop	Date		GI/wk[a]	No. insecticide applications	
		Planting	Harvest		FP	IPM
Song-jiang	Cabbage	2 Aug. 2000	8 Nov. 2000	0.49	2	1
Whenzou	Cabbage	16 Aug. 2000	25 Oct. 2000	0.85	4	2
Xiao-shan	Cabbage	7 Sept. 2000	20 Nov. 2000	0.63	3	1.8
Qiao-si	Cabbage	14 Sept. 2000	28 Nov. 2000	0.64	2	1.33
Ningbo	Cabbage	17 Sept. 2000	1 Dec. 2000	0.8	4	2
Song-jiang	Cabbage	1 Jan. 2000	1 May 2000	0.47	2	0.67
Song-jiang	Cauliflower	24 Aug. 2001	4 Nov. 2001	0.58	2.3	1
Whenzou	Cauliflower	30 Aug. 2001	15 Nov. 2001	0.57	2.3	1.67
Xiao-shan	Broccoli	7 Sept. 2001	20 Nov. 2001	0.58	2	1.67
Qiao-si	Broccoli	16 Sept. 2001	28 Nov. 2001	0.52	1.3	1.33
Song-jiang	Cauliflower	17 Sept. 2001	1 Dec. 2001	0.5	3	1.67

[a] For each trial the diamondback moth GI accumulated each week during the period in which the crop was growing was computed for use in regressions (see Materials and Methods).

was assumed that all diamondback moth populations have uniform responses to climatic variables. Most of the temperature related parameters were derived from Liu et al. (2002). Zalucki and Furlong (2011) refined the initial model by adjusting the moisture-related parameters, essentially making the species more dry tolerant, requiring concomitant adjustments in the dry and wet stress thresholds and similar changes for the degree-day accumulation of cold and heat stress parameters (Zalucki and Furlong 2008, 2011). The model was subsequently tested using diamondback moth damage data from China and areas predicted to support persistent diamondback moth populations correlated well with regions reporting the most severe pest damage (Li et al. 2012).

We present further independent tests of the diamondback moth CLIMEX model by generating seasonal phenologies of the pest at several locations. These were chosen as data were available, principal sources were the proceedings of the diamondback moth and other crucifer pest workshops (Talekar and Griggs 1986, Talekar 1992, Sivapragasam et al. 1997, Endersby and Ridland 2004, Shelton et al. 2008, Srinivasan et al. 2011). The seasonal phenologies were derived from data read off graphs or taken from tables that estimate the seasonal prevalence of diamondback moth based on counts of immatures in fields or adults in pheromone or light traps (see sources in Fig. 2). We have expressed the data as the percentage that occurred at each sampling date over a year. These seasonal phenologies use data that were not used to construct the model and therefore represent independent validation. The predictions are based on the standard CLIMEX output of weekly GI that indicates the suitability of a time period for diamondback moth populations, again expressed as a percentage of the total GI. We are only interested in the general visual concurrence of these graphs, as we are not attempting to make specific predictions of abundance. We present these tests to further justify the use of the diamondback moth-CLIMEX model as a reasonable approximation to the distribution and abundance of diamondback moth, an essential component of the method that we have developed to estimate how much it costs to manage this pest.

Estimating the Costs of Diamondback Moth Control. We establish a relationship between GI for a site and the application of insecticides for diamondback moth management by using data from several locations in the Changjiang River Valley, China, in 2000 and 2001. At each location diamondback moth abundance was assessed over one to two cropping seasons in experimental cabbage (*Brassica oleracea* Capitata), cauliflower (*B. oleracea* Botrytis), and broccoli (*B. oleracea* Italica) crops. Within these crops, the number of insecticide applications required using farmer practice (FP) or a threshold-based integrated pest management (IPM) approach was recorded (see Table 1 for details). We pool the data to establish the following broad relationships for insecticide applications under each management practice.

For FP plots or fields, no. of applications = 5.6 × GI/wk − 0.82 ($F_{1,9} = 16.25$; $P = 0.003$; $r^2_{adj} = 0.6$). For IPM plots or fields, no. of applications = 2.5 × GI/wk − 0.06 ($F_{1,9} = 9.84$; $P = 0.012$; $r^2_{adj} = 0.45$), where GI/wk is the average GI over the period the crop was grown at each location, using climate data from a nearby location.

In one set of calculations, we assume spray decision making for *Brassica* vegetable crops is based on GI as detailed above. For canola, we assume that crops are either treated with insecticide once per season or that crops are treated with insecticide after application of the same GI-based decision making processes that are applied to vegetable crops. In tropical countries, it is not uncommon for *Brassica* crops to be sprayed weekly or biweekly (Rauf et al. 2004, Sandur 2004, Mazlan and Mumford 2005); in our calculations, we use a single insecticide application per week as the worst-practice scenario.

We use annual world production statistics (FAOSTAT 2012) as the basis of calculations. Data are available for three categories of *Brassica* crops: cabbage and other brassicas (mainly leafy Chinese green vegetables), cauliflower and broccoli, and canola or oilseed rape. The production area represents the total

Table 2. Estimated annual costs of diamond back moth control for different crops on each continent when treated with insecticide weekly or managed by rational FP or an IPM strategy

Brassica crop	Continent	Cost of insecticide application strategy (US$)			
		Weekly	Single FP	Rational FP	IPM
Cabbage	Africa	$46,097,772		$19,202,651	$11,141,005
	Australia and Pacific	$1,169,364		$457,579	$267,545
	Asia	$695,435,398		$276,309,868	$161,195,709
	Europe	$216,137,670		$91,065,120	$52,762,140
	North and Central America	$42,129,738		$19,060,394	$10,952,758
	South America	$5,644,128		$2,231,623	$1,303,107
	Global total	$1,006,614,070		$408,327,234	$237,622,266
Cauliflower and broccoli	Africa	$5,433,960		$1,773,879	$1,063,439
	Australia and Pacific	$3,447,234		$1,292,478	$759,913
	Asia	$291,613,130		$111,784,037	$65,524,683
	Europe	$58,007,430		$23,238,373	$13,547,207
	North and Central America	$19,788,426		$8,361,502	$4,842,898
	South America	$23,656,962		$12,056,249	$6,840,740
	Global total	$401,947,142		$158,506,518	$92,578,880
Canola	Africa		$2,857,157	$13,475,282	$7,874,566
	Australia and Pacific		$45,818,497	$199,996,489	$118,066,148
	Asia		$475,528,620	$2,184,449,699	$1,280,853,119
	Europe		$211,162,386	$1,108,960,398	$639,659,862
	North and Central America		$195,127,090	$974,026,206	$565,206,806
	South America		$4,127,208	$23,057,347	$13,207,633
	Global total		$934,620,957	$4,503,965,422	$2,624,868,134

area of each crop grown in a country per year, and we calculated the mean production area of each crop for each country for which data are available from 2000 to 2009, the most up to date data available (FAOSTAT 2012). We assume that all crops take 12 wk from transplanting or sowing to harvest.

We also assume that if a region is suitable for diamondback moth (GI > 15), then we can distribute the crops proportionally to where diamondback moth is located and calculate the insecticide costs for each grid square. CLIMEX outputs a file of georeferenced data that shows the value of selected output variables for each x-y location or cell on a 30-min grid. We output EI, GI, weeks GI positive, degree-days, and number of diamondback moth generations for each location. We ported these data to ArcView (ESRI 2011) and used country shape files to distribute diamondback moth distribution and abundance by country. This enabled us to link country production area data for each of the crops to CLIMEX output that could then be linked to likely control costs based on the adoption of FP, IPM, or weekly applications of insecticide to various *Brassica* vegetable crops. Specifically, we distribute the total area of *Brassica* crop production to each grid location, j for example, relative to the GI value, for grid locations or cells with GI > 15, to get A_j.

Insecticide application costs can vary a great deal, and we use US$35/ha for farmer practice and US$40 for IPM for all crops. Pesticides vary enormously in costs depending on the product being used and application costs (e.g., fuel, labor costs). The higher cost for IPM reflects the additional labor costs of sampling and probably use of more expensive products. If all of these costs are included, then our estimates are not unreasonable for costs encountered in China (Liu et al. 2004), Canada (insecticide application to canola costs US$35/ha; L. Dosdall, personal communication) and the United States (insecticide and application costs estimated to be US$74/ha in U.S. vegetable crops; A. Shelton, personal communication).

Thus, the cost of management for each country will be given by the product of: $A_j \times SPH_j \times CPS_j$, summed over each grid location j, where A_j is the area of production in ha for each *Brassica* crop type, SPH_j is the number of insecticide applications per hectare either for the worst-case scenario (weekly), best FP or threshold-based IPM, and CPS_j is cost/spray for each crop. For simplicity of presentation we present data summed for each continent (Table 2); national area of production data and estimated control costs under the different application regimes can be found in Appendix 1.

The overall cost of diamondback moth will equal the cost of management plus the lost production even with management, i.e., yield loss with management. This is a measure of how ineffective management is due to poor spray timing, poor application (coverage), insecticide resistance, and so forth. Estimates regarding the yield losses caused by diamondback moth cover an enormous range and losses in many tropical developing countries are likely to be much higher than in temperate regions of industrialized countries. We use production values for the crops published by the Food and Agriculture Organization (FAO) (FAOSTAT 2012) and use these gross data and an assumed level of 5% decrease in crop value to estimate yield losses due to diamondback moth.

Test of CLIMEX Model

Predicted and Actual Geographic Distributions. The predicted worldwide distribution, mapped as EI values (Fig. 1a) and GI values (Fig. 1b) based on the set of CLIMEX parameters described in Zalucki and Furlong (2011) for diamondback moth agrees well

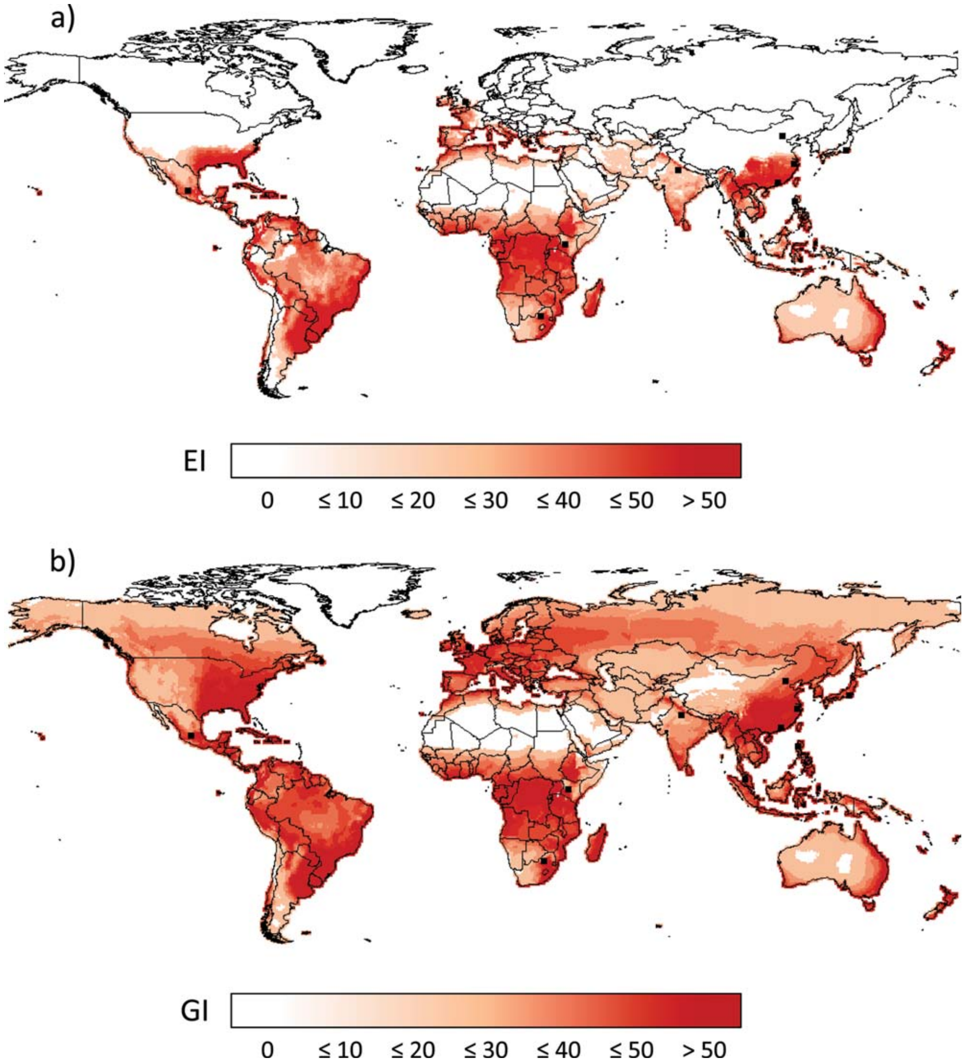

Fig. 1. Predicted geographic distribution of diamondback moth worldwide. (a) Values where EI is positive denote the core of the species range and year round occupation. (b) Regions where GI is positive indicate the potential migratory seasonal range. The "actual" distribution CAB maps can be found in the CABI Crop Protection Compendium (CABI 2012).

with the "known" worldwide distribution (Commonwealth Institute of Entomology 1967, CAB 2012). Regions with values of EI below ≈15 are probably marginal for long-term persistence of a species; essentially such conditions are associated with the edge of the species core range (Fig. 1a). The potential increase in the species range when GI values are plotted (Fig. 1b) should be noted. This area represents the maximal range that diamondback moth can exploit if it migrates out of its core (Fig. 1a) and colonizes these locations at a time when prevailing seasonal conditions result in a positive GI. Such migrations are highly variable from year to year (Zalucki and Furlong 2005, 2008, 2011).

The fine scale–predicted distribution of diamondback moth in Japan (Fig. 1a and b) agrees well with observations of the species' actual distribution in this region. Diamondback moth does not persist in northern Japan, but it does so in southern regions (Honda 1992) and makes annual reinvasions of northern Japan. Again the geographic distribution in Japan is a prediction based on our model parameter values, and it was not used to generate the model. Similarly, the species does not usually persist on the Korean peninsula but reinvades each year, most likely from China, and populations build up when GI values are suitable in the spring–summer (Furlong et al. 2008). The northern range limit of seasonal breeding in China (Fig. 1a) seems to agree well with observations there (Li et al. 2012).

Predicted and Actual Seasonal Phonologies. CLIMEX can be used to predict average season phenology at a site in terms of weekly growth indices. We have done so for 11 locations around the world that vary widely in their suitability for diamondback moth, by using climatic data for an average year at each location (locations are marked on Fig. 1). Again, the predicted values agree well with observed values across a range of sites (Fig. 2a–k). This is not unexpected for the Cameron Highlands, Malaysia (Fig. 2a) because these data were used to parameterize the initial model (Zalucki and Furlong 2008). However, many other sites across a wide range of geographic locations(northern Luzon, Philippines [Fig. 2b], China [Fig. 2c–e; in Fig. 2e, high populations recorded in weeks 21–27 probably are the result of migration event; population changes after week 28 follow a very similar pattern to the growth indices predicted by the model], Japan [Fig. 2f], India [Fig. 2g], Kenya [Fig. 2h], the United Kingdom [Fig. 2i], and Mexico [Fig. 2j]) show good visual agreement between observed and predicted phenology, but there is a poor fit for a site in South Africa (Fig. 2k). The sites that were used for these comparisons were sites for which diamondback moth population monitoring data could be identified in one form or another (sampling of larvae on plants, pheromone traps, and light traps) to estimate the species seasonal phenology/abundance at the site. Importantly, these abundance data reflect not only vital population rates of birth, death, and movement as influenced by climate and host plant (crop and noncrop weedy *Brassica* species) but also widespread intensive management, namely, insecticide application. In general the model's GI is a leading indicator of population buildup, as one would expect as it is a measure of potential growth rate. Outside areas where EI is positive (Fig. 1a), the seasonal dynamics will have a strong migration component. That our simple model by and large captures both spatial (Fig. 1a and b) and seasonal temporal dynamics (Fig. 2) argues that there is nothing particularly special about the species' biology and ecology at specific locations throughout its range. The predictions are based on one set of parameters and climate data for each location. We therefore use our CLIMEX model to "predict" diamondback moth abundance throughout its range and undertake an analysis of possible management costs.

Estimated Cost of Diamondback Moth Management and Losses

A few things stand out in terms of how much diamondback moth control costs the world economy (Table 2). For vegetable crops, if management is based on weekly spray regimes, then diamondback moth costs ≈US$1.4 billion. If diamondback moth is only treated with insecticide as required, when average GI is suitable for diamondback moth population growth, then the predicted control costs are ≈US$566 million; this drops to ≈US$340 million if IPM strategies are assumed to be universally adopted for vegetable crops (Table 2). The latter assumes diamondback moth abundance is entirely determined by climate. We have not calculated a control cost based on weekly spraying of canola because this is unlikely ever to be the case. Spraying this crop a single time during the crop cycle adds ≈US$935 million to management costs if a single spray farmer practice intervention is assumed (Table 2). If diamondback moth in canola is managed using IPM or farmer practice regimes based on climate that are equivalent to those in vegetable crops, then control costs are estimated to be in the region of US$2.6 billion and US$4.5 billion, respectively (Table 2). However, such assessments must be interpreted with caution, as in canola economic thresholds are usually much higher than for *Brassica* vegetable crops.

Based on these estimates our approach predicts a lower bound for the management costs of diamondback moth in all crops of ≈US$1.27 billion (US$340 million for all vegetables, assuming the adoption of IPM, and US$935 million for canola, assuming a single insecticide application) and an upper bound of ≈US$2.3 billion (US$1.4 billion for all vegetables based on weekly insecticide application and US$935 million for canola, assuming a single insecticide application across the global distribution of the crop). The real value will lie somewhere between these bounds. In 2009, the annual value of *Brassica* vegetable and canola crops was estimated to be >US$51 billion (FAO STAT 2012), meaning that just 5% crop loss due to the damage caused by diamondback moth would add another US$2.7 billion to the total economic costs, which can be attributed to this destructive pest. So, the piece of string could be US$5.0 billion long!

Vegetable production is extremely high in Asia and this continent consequently dominates loss estimates (Table 2; Fig. 3). Examination of Fig. 3 allows calculation of estimated control costs for a given continent based on the assumption of a continuum (0–100%) of weekly insecticide applications; when crops are not treated weekly it is assumed that there is equal adoption of farmer practice and IPM-based management strategies. Using these data it is possible to determine a range of estimates for the worldwide costs of diamondback moth management by assuming different adoption rates of different pest management strategies in different continents. Such an approach will more likely lead to a more precise estimate of actual management costs than we attempt above. Similarly the national data that were generated and used to determine the continental estimates (Appendix 1) are a useful resource that enables diamondback moth control costs in each country to be estimated based on assumptions regarding the local adoption rates of different pest management strategies.

Discussion

If we assume that the cost of diamondback moth of US$1billion was correct in 1993, then, at a minimum, one should allow for inflation. The net present value of this sum is US$2.65 billion assuming 5% inflation per annum and US$1.81 billion if an inflation rate of 3% is applied. Furthermore, there have been dramatic increases in both *Brassica* vegetable (39% increase) and canola (59% increase) production areas since 1993, and there is now an additional crop area of >12 million ha/annum available to the pest that requires protection (FAOSTAT 2012). This represents an overall in-

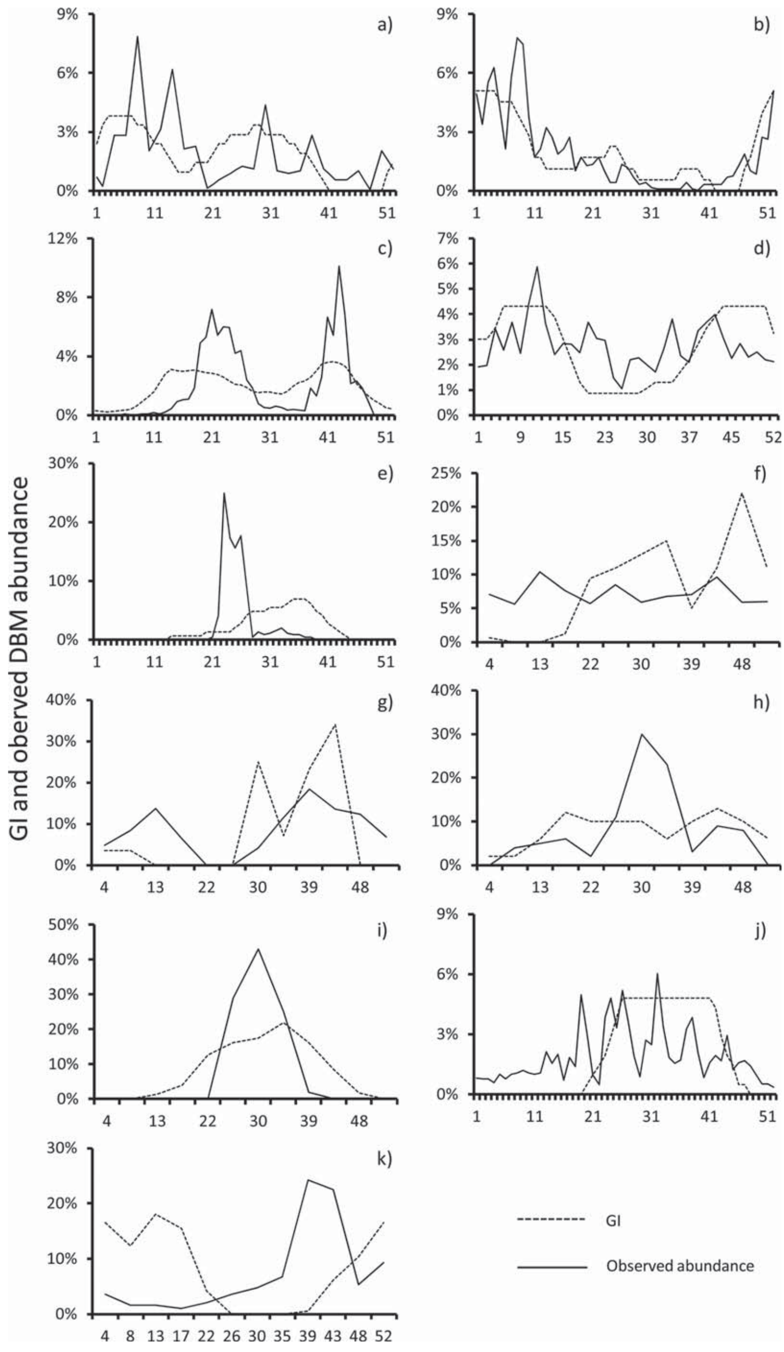

Fig. 2. Seasonal phenology of diamondback moth for locations throughout its range (see Fig. 1) based on various methods to observe the species abundance (solid line, details below) plotted with the **CLIMEX GI** value (dashed line) based on average climatic conditions at the location provided by **CLIMEX**. The x-axis varies depending on how the base data were collected (e.g., weekly, monthly). (a) Cameron highlands, Malaysia, (4° 28′ N, 101° 23′ E; 1,470 m), larvae per 10 plants (Ooi 1992). (b) La Trinidad, northern Luzon (8° 56′ N, 125° 31′ E), light and pheromone trap (Poelking 1992). (c) Hangzhou, China (30° 14′ N, 120° 26′ E) (Zalucki and Furlong 2011). (d) Huadu, China (23° 23′ N, 114° 28′ E; 6.6 m) (Li et al. 2012). (e) Beijing (40° 28′ N, 115° 58′ E), China, larvae per 100 plants (Shi et al. 2008). (f) Hiratsuka, Kanagawa (35° 19′ N, 139° 21′ E), Japan, pheromone trap (Kuwahara et al. 1995). (g) Aligarh district (27° 53′ N, 78° 5′ E), India, larvae per plant, (Ahmad and Ansari 2010). (h) Naivasha, Nakuru district, the Rift Valley, altitude, 1,500 m (00° 45′ S, 036° 26′ E), Kenya, larvae per 20 plants (Rossbach et al. 2008). (i) Holbeach, Lincolnshire (52° 48′ N, 0° 1′ E), United Kingdom, larvae per 20 plants (Collier and Finch 2004). (j) Salamanca (20° 34′ N, 101° 12′ W), Mexico, pheromone trap (McCully and Araiza Salas 1992). (k) Brits (25° 25′ S, 27° 76′ E, altitude, 1,100 m), South Africa, pheromone trap (Kfir 1997, 2004; Nofemela and Kfir 2008).

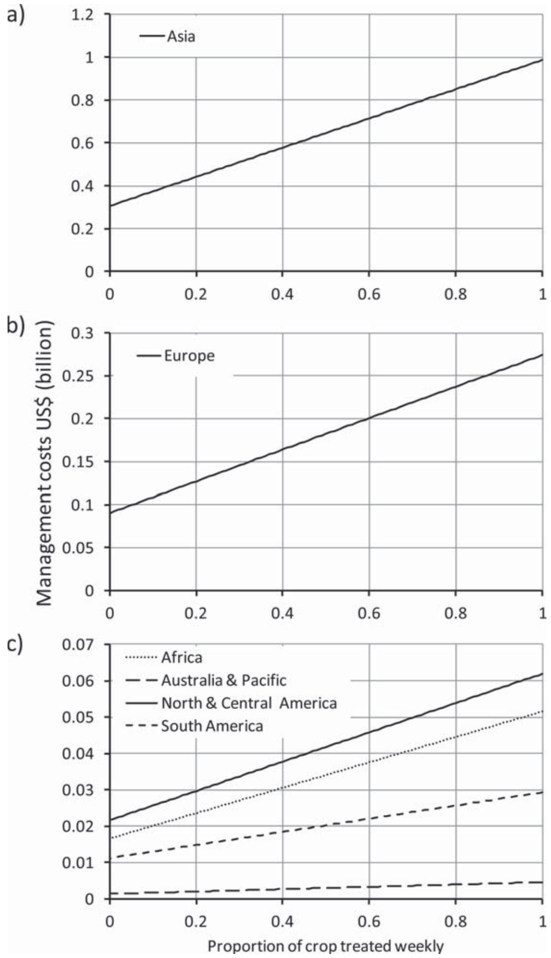

Fig. 3. Estimated costs of diamondback moth control in vegetable *Brassica* crops for each continent (cabbages [and leafy Chinese greens], cauliflower and broccoli) along a continuum of insecticide interventions: 1, weekly application of insecticide by 100% of farmers in a given region; 0, weekly application of insecticide by 0% of farmers in a given region. When crops are not treated with insecticide weekly, equal adoption of FP- and IPM-based insecticide applications are assumed.

crease of 57% to the global *Brassica* cropping area since 1993, and whatever the merits of the previous US$1 billion estimate it is clearly outdated, even without consideration of the increased intensification of practices that have accompanied this change.

Our analysis suggests that the management of diamondback moth in *Brassica* vegetable crops alone costs US$1.4 billion when based on the conservative assumption of a management regime of one insecticide application per week (Table 2). When the cost of managing diamondback moth in canola is included the total cost estimate for management increases to US$2.3 billion. These estimates are at the extreme range and represent the worst-case scenario of a single spray to every canola crop and weekly spraying for vegetables (Table 2). The latter is not uncommon for diamondback moth on vegetable crops in many parts of the species' range, particularly in the tropics where insecticide application rates can be even higher (Mazlan and Mumford 2005, Williamson 2005, Atumurirava and Furlong 2011). As has been suggested by many studies, diamondback moth is an induced pest that can, to a large extent, be controlled by natural enemies (Kfir 2004, Furlong et al. 2008, Li et al. 2012).

Our work in China (Liu et al. 2005), Democratic People's Republic of Korea (Furlong et al. 2008), and Australia (Furlong et al. 2004a,b) indicates that it is possible to greatly reduce management costs by moving away from weekly insecticide application and the adoption of threshold-based IPM (Zalucki et al. 2009). If management was based on diamondback moth populations requiring control only when climatic conditions were suitable, then even rational farmer practice would reduce costs to US$560 million in vegetables and adoption of IPM would reduce these costs by a further 42% (Table 2). As has been found in Australia (Furlong et al. 2004a,b); China (Li et al. 2012); the Cameron Highlands, Malaysia (Ooi 1992); New Zealand (Walker et al. 2004); and South Africa (Kfir 1997, 2004), if natural enemies are not disrupted by broad-spectrum insecticides and allowed to exert population suppressive forces on the pest, then costs associated with the management diamondback moth would be even less.

The cost estimate is based on a CLIMEX prediction of population size based on GI and decision making as in our trials in China (Table 2). So, how reasonable is our model and approach? CLIMEX has been used extensively in biological control programs and pest risk analysis to predict potential distributions of various species (for brief reviews, see Zalucki and Furlong 2005, van Klinken et al. 2009, Lawson et al. 2010). However, this method has not yet been used widely to model population changes of a species (but see Zalucki and Furlong 2005; Zalucki and van Klinken 2006; Zalucki and Furlong 2008, 2011; Li et al. 2012). The CLIMEX parameter set integrates data on species distribution into a single model. This approach may be used to generate a climate driven null model for the abundance of a species at a site over time. Such models that enable one to infer likely species abundance based on climate are critical to testing the effectiveness of management strategies such as areawide management and the planting of transgenic crops (Carrière et al. 2003, Zalucki and Furlong 2005), as well as predicting temporal dynamics and outbreaks of pests (e.g., Maelzer et al. 1996). Historical climate data can be used to predict both the expected changes in range and variation in the seasonal and annual temporal dynamics of the pest and therefore of likely pest status.

In our estimates of management costs, we use our CLIMEX model to generate diamondback moth spatial abundance and the scale of the problem that it represents. The validity of the model rests on how well the predictions match reality. Geographically, the predictions accord well with notions about the distribution of diamondback moth, but surprisingly, there is no real data to test against per se. Predictions of seasonal phenology (Fig. 2) provide good approximations to field data for almost all locations tested. Zalucki and Furlong (2008, 2011) and Li et al. (2012) took this one step further and showed that the model was even useful for long-term population data sets.

Our other assumption is that *Brassica* crops are the same as diamondback moth, and we have distributed *Brassica* production areas accordingly. In practice, areas of agriculture are more concentrated depending on soil, topography, water availability, and other factors that affect land use. The distribution of host plants also will impact on populations of insects (e.g., Zalucki and Lammers 2010) and of course their pest status (Schellhorn et al. 2008). These matters are beyond the scope of this paper, but they are unlikely to change key findings substantially. We might expect areas of high production to lead to higher levels of pest abundance—the resource concentration hypothesis on a landscape scale.

Our country estimates are close to local experts opinions. In the United States, Anthony Shelton (personal communication) estimates that management cost associated with *Brassica* vegetables are of the order of US$220 million; our equivalent estimate is ≈US$46 million. The difference can be explained due to different estimates of the area of production; the FAO estimates that the area of production is ≈71,000 ha (FAOSTAT 2012), whereas Shelton used USDA estimates of 107,000 ha; and differences in assumed management costs per ha; we use US$35–40/ha but Shelton uses more than double this figure (see above). Allowing for these differences, our equivalent estimate would be ≈US$150 million, a value that is surprisingly close. Sandur (2004) has estimated that diamondback moth management in vegetable crops in India costs ≈US$168 million/annum, we estimate costs to be US$45–US$217 million, depending on the relative degree of adoption of IPM and weekly insecticide application (Appendix 1).

Our estimates are probably conservative for two major reasons. First, our assumptions about management costs and the frequency of insecticide use is likely to be an underestimate of common practices in many parts of the world where insecticides are frequently applied to *Brassica* vegetable crops more than two times per week (Mazlan and Mumford 2005, Williamson 2005, Atumurirava and Furlong 2011). If this were universally the case, then the upper cost for vegetables would be double our current estimate. Second, the official FAO statistics that we have used as the basis of our cost calculations (FAOSTAT 2012) underestimate the area of *Brassica* crops grown. This is well illustrated by the example the United States discussed above, and Appendix 1 shows that the production area data that we used (FAOSTAT 2012) indicate that many countries, even some in which diamondback moth is a renowned and significant pest, e.g., Taiwan, Myanmar, Laos, and Papua New Guinea, produce no *Brassica* crops at all! Indeed, the database returns zero production data for 35 of the 168 countries for which data could be retrieved. Although the effect of such discrepancies is in itself significant, an even greater source of error is likely to derive from the fact that in many developing countries significant quantities of *Brassica* crops are grown in the subsistence sector. Although these crops are frequently intensively managed with insecticides, they often contribute only to local rural economies, and as such production data do not enter official statistics.

In conclusion, diamondback moth is an enormous economic problem, and certainly it now costs much >US$1 billion/annum to manage. Management costs and lost production probably make diamondback moth a US$4–US$5 billion problem. Current management relies heavily on insecticide applications, sometimes under the guise of threshold based IPM. As has happened repeatedly, insecticide resistance develops rapidly (e.g., Li et al. 2012) and reliance on insecticides is not sustainable. Our experience and analysis suggest that worldwide adoption of even basic IPM would dramatically reduce costs and allow natural enemies to contribute to pest population suppression. To achieve more efficient and long-term control, adoption of improved management must be facilitated; this is "déjà vu all over again" (Shelton 2004).

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Appendix 1. Mean production data and associated cost estimates for diamondback moth management under different management practices and scenarios for all countries for which data were available (FAOSTAT 2012)

Continent	Country	Mean production area (Ha) 2000–2009			Estimated costs (US$) based on farmer practice (FP)			Estimated costs (US$) based on IPM adoption			Estimated costs (US$) based on weekly application of insecticide (FP)		Estimated costs based of single application of insecticide to canola	
		Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	FP	IPM
Africa	Algeria	2,580	4,175	15,199	$297,585	$481,594	$1,753,243	$183,413	$296,825	$1,080,592	$1,083,516	$1,753,500	$531,969	$607,964
Africa	Angola	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Benin	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Botswana	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Burkina Faso	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Burundi	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Cameroon	2,874	0	0	$529,622	$0	$0	$305,397	$0	$0	$1,206,954	$0	$0	$0
Africa	Central African Republic	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Chad	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Congo	91	0	0	$19,432	$0	$0	$11,030	$0	$0	$38,262	$0	$0	$0
Africa	Côte d'Ivoire	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Democratic Republic of the Congo	1,516	0	0	$309,090	$0	$0	$176,258	$0	$0	$636,678	$0	$0	$0
Africa	Egypt	20,199	4,798	0	$2,745,866	$652,294	$0	$1,648,247	$391,549	$0	$8,483,622	$2,015,328	$0	$0
Africa	Equatorial Guinea	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Eritrea	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Ethiopia	21,308	0	24,150	$3,871,402	$0	$4,387,795	$2,236,076	$0	$2,534,338	$8,949,360	$0	$845,257	$966,008
Africa	Gabon	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Ghana	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Guinea	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Kenya	32,824	92	0	$6,685,930	$18,679	$0	$3,813,042	$10,653	$0	$13,785,870	$38,514	$0	$0
Africa	Lesotho	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Liberia	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Madagascar	788	53	0	$177,216	$11,982	$0	$100,067	$6,766	$0	$331,086	$22,386	$0	$0
Africa	Malawi	3,418	0	0	$631,586	$0	$0	$364,078	$0	$0	$1,435,350	$0	$0	$0
Africa	Mali	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Mauritania	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Morocco	1,462	2,066	940	$225,901	$319,346	$145,284	$133,150	$188,228	$85,633	$613,872	$867,804	$32,900	$37,600
Africa	Mozambique	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Namibia	145	0	0	$17,736	$0	$0	$10,824	$0	$0	$60,900	$0	$0	$0
Africa	Niger	9,192	0	0	$920,872	$0	$0	$582,367	$0	$0	$3,860,556	$0	$0	$0
Africa	Nigeria	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Rwanda	4,259	0	0	$1,221,522	$0	$0	$675,370	$0	$0	$1,788,864	$0	$0	$0
Africa	Senegal	1,642	0	0	$162,644	$0	$0	$103,085	$0	$0	$689,682	$0	$0	$0
Africa	Sierra Leone	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Somalia	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	South Africa	2,785	1,178	36,964	$496,851	$210,148	$6,594,660	$287,590	$121,639	$3,817,164	$1,169,658	$494,718	$1,293,731	$1,478,550
Africa	Sudan	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Swaziland	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Togo	44	0	0	$6,494	$0	$0	$3,856	$0	$0	$18,606	$0	$0	$0
Africa	Tunisia	1,556	544	4,380	$211,167	$73,813	$594,301	$126,792	$44,320	$356,838	$653,646	$228,480	$153,300	$175,200
Africa	Uganda	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	United Republic of Tanzania	3,047	0	0	$666,535	$0	$0	$377,377	$0	$0	$1,279,866	$0	$0	$0

Appendix 1. Continued

Continent	Country	Mean production area (Ha) 2000–2009			Estimated costs (US$) based on farmer practice (FP)			Estimated costs (US$) based on IPM adoption			Estimated costs (US$) based on weekly application of insecticide (FP)		Estimated costs based of single application of insecticide to canola	
		Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	FP	IPM
Africa	Western Sahara	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Zambia	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Africa	Zimbabwe	27	32	0	$5,202	$6,024	$0	$2,987	$3,459	$0	$11,424	$13,230	$0	$0
Asia	Afghanistan	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Asia	Azerbaijan	6,286	0	0	$512,233	$0	$0	$338,307	$0	$0	$2,640,288	$0	$0	$0
Asia	Bangladesh	13,606	13,049	266,274	$2,525,335	$2,421,901	$49,420,247	$1,455,018	$1,395,422	$28,474,369	$5,714,688	$5,480,622	$9,319,597	$10,650,968
Asia	Bhutan	698	158	0	$105,119	$23,811	$0	$62,180	$14,084	$0	$293,230	$66,420	$0	$0
Asia	Cambodia	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Asia	China	1,007,443	343,496	6,900,183	$184,176,369	$62,796,470	$1,261,461,219	$106,301,520	$36,244,390	$728,080,617	$423,126,186	$144,268,404	$241,506,405	$276,007,320
Asia	India	261,720	281,790	5,906,150	$36,148,477	$38,920,523	$815,750,900	$21,647,301	$23,307,324	$488,507,589	$109,922,400	$118,351,800	$206,715,250	$236,246,000
Asia	Indonesia	68,055	7,250	0	$8,339,115	$888,317	$0	$5,087,838	$541,978	$0	$28,583,100	$3,044,790	$0	$0
Asia	Iran	11,279	1,144	105,800	$1,278,847	$129,682	$11,996,457	$790,554	$80,166	$7,415,936	$4,736,970	$480,354	$3,703,000	$4,232,000
Asia	Iraq	1,275	2,370	0	$153,611	$285,536	$0	$93,983	$174,697	$0	$535,500	$995,400	$0	$0
Asia	Israel	1,974	1,474	0	$397,616	$296,863	$0	$227,033	$169,504	$0	$829,080	$618,996	$0	$0
Asia	Japan	46,630	12,041	728	$7,546,034	$1,948,570	$117,875	$4,420,902	$1,141,584	$69,058	$19,584,600	$5,057,220	$25,494	$29,136
Asia	Jordon	710	2,390	0	$130,474	$439,109	$0	$75,261	$253,290	$0	$298,200	$1,003,590	$0	$0
Asia	Kazakhstan	14,865	23	72,826	$1,441,387	$2,230	$7,061,737	$917,387	$1,419	$4,494,521	$6,243,132	$9,660	$2,548,896	$2,913,024
Asia	Korea North	34,474	0	0	$6,124,644	$0	$0	$3,546,879	$0	$0	$14,479,080	$0	$0	$0
Asia	Korea South	48,404	0	1,045	$8,433,009	$0	$182,008	$4,895,164	$0	$105,651	$20,329,680	$0	$36,575	$41,800
Asia	Kuwait	209	232	0	$0	$0	$0	$0	$0	$0	$87,906	$97,524	$0	$0
Asia	Kyrgyzstan	5,017	117	486	$423,294	$9,872	$41,041	$277,385	$6,469	$26,894	$2,107,014	$49,140	$17,024	$19,456
Asia	Laos	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Asia	Lebanon	1,563	1,015	0	$278,047	$180,502	$0	$160,995	$104,514	$0	$656,418	$426,132	$0	$0
Asia	Malaysia	1,508	0	0	$216,141	$0	$0	$128,738	$0	$0	$633,360	$0	$0	$0
Asia	Mongolia	1,652	0	0	$189,773	$0	$0	$117,051	$0	$0	$693,924	$0	$0	$0
Asia	Myanmar	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Asia	Nepal	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Asia	Oman	296	132	0	$11,633	$5,175	$0	$9,559	$4,252	$0	$124,320	$55,300	$0	$0
Asia	Pakistan	4,431	11,389	322,738	$512,343	$1,316,988	$37,321,360	$315,641	$811,363	$22,992,742	$1,860,810	$4,783,254	$11,295,820	$12,909,508
Asia	Philippines	7,899	1,065	0	$1,440,777	$194,286	$0	$831,797	$112,166	$0	$3,317,580	$447,370	$0	$0
Asia	Qatar	97	90	0	$0	$0	$0	$0	$0	$0	$40,740	$37,926	$0	$0
Asia	Saudi Arabia	2,798	0	0	$120,073	$0	$0	$95,516	$0	$0	$1,175,118	$0	$0	$0
Asia	Sri Lanka	4,003	0	0	$742,953	$0	$0	$428,066	$0	$0	$1,681,260	$0	$0	$0
Asia	Syrian Arab Republic	2,010	1,654	0	$242,378	$199,399	$0	$148,269	$121,978	$0	$844,158	$694,470	$0	$0
Asia	Taiwan	0	0	0	$0			$0			$0	$0	$0	$0
Asia	Tajikistan	1,697	0	432	$149,624	$0	$38,050	$97,113	$0	$24,696	$712,656	$0	$15,103	$17,260
Asia	Thailand	27,331	5,251	0	$3,549,605	$682,011	$0	$2,145,627	$412,254	$0	$11,478,810	$2,205,504	$0	$0
Asia	Turkey	29,434	5,979	8,213	$3,314,594	$673,335	$924,830	$2,051,475	$416,742	$572,397	$12,362,280	$2,511,306	$287,441	$328,504
Asia	Turkmenistan	3,030	0	0	$0	$0	$0	$0	$0	$0	$1,272,600	$0	$0	$0
Asia	United Arab Emirates	782	365	0	$0	$0	$0	$0	$0	$0	$328,566	$153,132	$0	$0
Asia	Uzbekistan	8,983	0	1,658	$726,052	$0	$133,976	$480,412	$0	$88,649	$3,772,860	$0	$58,016	$66,304
Asia	Viet Nam	35,169	1,845	0	$7,043,179	$369,457	$0	$4,024,021	$211,084	$0	$14,770,770	$774,816	$0	$0
Asia	Yemen	472	0	0	$37,130	$0	$0	$24,719	$0	$0	$198,114	$0	$0	$0
Australia & Pacific	Australia	1,929	7,105	1,307,171	$294,555	$1,085,090	$199,633,716	$173,896	$640,603	$117,857,444	$810,054	$2,984,100	$45,750,985	$52,286,840
Australia & Pacific	Fiji	70	0	0	$15,255	$0	$0	$8,638	$0	$0	$29,316	$0	$0	$0
Australia & Pacific	New Caledonia	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Australia & Pacific	New Zealand	786	1,103	1,929	$147,769	$207,388	$362,774	$85,011	$119,310	$208,704	$329,994	$463,134	$67,512	$77,156
Australia & Pacific	Papua New Guinea	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Australia & Pacific	Solomon Islands	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Australia & Pacific	Vanuatu	0	0	0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0

Appendix 1. Continued

Continent	Country	Mean production area (Ha) 2000–2009			Estimated costs (US$) based on farmer practice (FP)			Estimated costs (US$) based on IPM adoption			Estimated costs (US$) based on weekly application of insecticide (FP)		Estimated costs based of single application of insecticide to canola	
		Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	FP	IPM
Europe	Albania	1,291	161	0	$203,491	$25,419	$0	$104,676	$13,075	$0	$542,346	$67,746	$0	$0
Europe	Armenia	3,328	247	0	$313,272	$23,235	$0	$200,571	$14,876	$0	$1,397,550	$103,656	$0	$0
Europe	Austria	1,826	296	48,189	$351,379	$57,033	$9,271,598	$201,634	$32,727	$5,320,382	$767,046	$124,500	$1,686,626	$1,927,572
Europe	Belarus	21,225	0	154,093	$4,447,894	$0	$32,291,256	$2,529,191	$0	$18,361,667	$8,914,584	$0	$5,393,248	$6,163,712
Europe	Belgium	3,711	4,035	6,956	$759,397	$825,736	$1,423,399	$432,883	$470,699	$811,388	$1,558,704	$1,694,868	$243,467	$278,248
Europe	Bosnia and Herzegovina	6,569	0	811	$1,359,386	$0	$167,766	$773,988	$0	$95,520	$2,758,980	$0	$28,375	$32,428
Europe	Bulgaria	4,401	334	33,926	$581,787	$44,211	$4,485,270	$350,705	$26,651	$2,703,748	$1,848,210	$140,448	$1,187,393	$1,357,020
Europe	Croatia	5,771	224	15,880	$1,208,497	$46,949	$3,325,303	$687,234	$26,699	$1,890,994	$2,423,820	$94,164	$555,783	$635,180
Europe	Cyprus	124	77	0	$22,078	$13,645	$0	$12,783	$7,901	$0	$52,122	$32,214	$0	$0
Europe	Czech Republic	3,150	972	309,816	$641,590	$198,063	$63,111,117	$365,902	$112,956	$35,992,581	$1,322,832	$408,366	$10,843,557	$12,392,636
Europe	Denmark	828	608	124,219	$173,546	$127,534	$26,039,115	$98,680	$72,517	$14,806,062	$347,718	$255,528	$4,347,676	$4,968,772
Europe	Estonia	819	62	52,878	$167,102	$12,583	$10,783,603	$95,288	$7,175	$6,149,221	$344,148	$25,914	$1,850,744	$2,115,136
Europe	Finland	1,015	517	75,410	$155,276	$79,083	$11,535,150	$91,650	$46,678	$6,808,515	$426,342	$217,140	$2,639,350	$3,016,400
Europe	France	9,793	26,864	1,267,009	$1,706,918	$4,682,328	$220,834,869	$990,773	$2,717,838	$128,182,662	$4,113,144	$11,282,964	$44,345,315	$50,680,360
Europe	Georgia	10,660	0	0	$1,982,314	$0	$0	$1,141,894	$0	$0	$4,477,200	$0	$0	$0
Europe	Germany	15,065	6,560	1,322,457	$3,016,672	$1,313,528	$264,815,506	$1,723,556	$750,475	$151,300,604	$6,327,258	$2,755,032	$46,285,995	$52,898,280
Europe	Greece	8,417	4,381	3,750	$1,117,811	$581,834	$498,021	$673,359	$350,491	$300,003	$3,535,098	$1,840,062	$131,250	$150,000
Europe	Hungary	5,067	1,266	152,780	$752,301	$187,949	$22,685,084	$445,857	$111,389	$13,444,481	$2,127,972	$531,636	$5,347,286	$6,111,184
Europe	Ireland	899	909	4,200	$120,351	$121,690	$562,449	$72,406	$73,212	$338,384	$377,454	$381,654	$147,000	$168,000
Europe	Italy	20,232	20,817	13,533	$3,632,788	$3,737,831	$2,430,005	$2,101,156	$2,161,911	$1,405,482	$8,497,272	$8,742,972	$473,659	$541,324
Europe	Latvia	3,108	227	54,360	$679,212	$49,692	$11,878,886	$384,590	$28,137	$6,726,178	$1,305,444	$95,508	$1,902,600	$2,174,400
Europe	Lithuania	5,238	353	112,150	$1,134,463	$76,370	$24,290,745	$642,933	$43,281	$13,766,274	$2,199,876	$148,092	$3,925,250	$4,486,000
Europe	Netherlands	7,189	3,850	1,836	$1,428,425	$765,011	$364,861	$816,799	$437,447	$208,634	$3,019,254	$1,617,000	$64,267	$73,448
Europe	Norway	1,029	947	7,053	$156,170	$143,640	$1,070,022	$80,746	$74,267	$553,243	$432,180	$397,740	$246,855	$282,120
Europe	Poland	37,501	13,493	583,532	$7,490,246	$2,695,071	$116,551,114	$4,280,674	$1,540,233	$66,608,948	$15,750,462	$5,667,186	$20,423,606	$23,341,264
Europe	Portugal	9,353	2,071	0	$1,739,614	$385,110	$0	$1,002,069	$221,835	$0	$3,928,386	$869,652	$0	$0
Europe	Republic of Moldova	4,458	118	16,302	$620,152	$16,414	$2,267,548	$370,987	$9,819	$1,356,489	$1,872,486	$49,560	$570,553	$652,060
Europe	Romania	44,900	1,536	153,127	$7,869,272	$269,235	$26,837,204	$4,564,644	$156,172	$15,567,168	$18,858,168	$645,204	$5,359,459	$6,125,096
Europe	Russian Federation	147,208	1,267	322,734	$25,638,954	$220,723	$56,210,003	$14,883,345	$128,129	$32,629,759	$61,827,360	$532,266	$11,295,690	$12,909,360
Europe	Serbia and Montenegro	23,636	0	3,461	$4,323,296	$0	$633,040	$2,495,129	$0	$365,350	$9,926,910	$0	$121,129	$138,433
Europe	Slovakia	6,457	1,090	117,629	$1,242,109	$209,660	$22,627,814	$712,781	$120,313	$12,984,916	$2,711,940	$457,758	$4,117,008	$4,705,152
Europe	Slovenia	836	89	2,690	$168,805	$18,040	$543,345	$96,355	$10,297	$310,145	$350,952	$37,506	$94,136	$107,584
Europe	Spain	8,474	24,660	12,555	$1,086,725	$3,162,352	$1,610,024	$658,198	$1,915,347	$975,146	$3,559,122	$10,356,990	$439,415	$502,188
Europe	Sweden	414	418	73,456	$69,549	$70,136	$12,328,141	$40,558	$40,900	$7,189,179	$174,048	$175,518	$2,570,960	$2,938,240
Europe	Switzerland	1,081	658	17,308	$198,458	$120,816	$3,178,325	$114,485	$69,695	$1,833,488	$453,894	$276,318	$605,763	$692,300
Europe	The former Yugoslav Republic of Macedonia	3,523	24	698	$423,947	$2,840	$84,029	$259,426	$1,738	$51,420	$1,479,492	$9,912	$24,437	$27,928
Europe	Ukraine	74,930	1,770	427,170	$12,366,548	$292,123	$70,500,710	$7,226,820	$170,712	$41,199,531	$31,470,600	$743,400	$14,950,950	$17,086,800
Europe	United Kingdom	11,089	17,212	541,245	$1,715,323	$2,662,488	$83,725,078	$1,010,923	$1,569,134	$49,343,266	$4,657,296	$7,228,956	$18,943,589	$21,649,816
North & Central America	Belize	60	0	0	$11,500	$0	$0	$6,601	$0	$0	$25,158	$0	$0	$0
North & Central America	Canada	8,647	2,312	5,116,870	$1,494,696	$399,699	$884,528,363	$868,459	$232,236	$513,935,089	$3,631,572	$971,124	$179,090,450	$204,674,800

Appendix 1. Continued

Continent	Country	Mean production area (Ha) 2000–2009			Estimated costs (US$) based on farmer practice (FP)			Estimated costs (US$) based on IPM adoption			Estimated costs (US$) based on weekly application of insecticide (FP)		Estimated costs based of single application of insecticide to canola	
		Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	Canola	Cabbages & other brassicas	Cauliflower and broccoli	FP	IPM
North & Central America	Costa Rica	2,997	0	0	$425,580	$0	$0	$253,821	$0	$0	$1,258,614	$0	$0	$0
North & Central America	Cuba	8,302	0	0	$1,677,686	$0	$0	$957,607	$0	$0	$3,487,008	$0	$0	$0
North & Central America	Dominican republic	362	0	0	$96,235	$0	$0	$53,535	$0	$0	$152,166	$0	$0	$0
North & Central America	El salvador	89	0	0	$9,261	$0	$0	$5,815	$0	$0	$37,380	$0	$0	$0
North & Central America	Guatemala	3,898	5,108	0	$818,173	$1,072,222	$0	$465,152	$609,585	$0	$1,636,950	$2,145,234	$0	$0
North & Central America	Haiti	1,710	0	0	$401,120	$0	$0	$225,586	$0	$0	$718,116	$0	$0	$0
North & Central America	Honduras	1,837	107	0	$365,766	$21,280	$0	$209,110	$12,166	$0	$771,708	$44,898	$0	$0
North & Central America	Jamaica	1,288	119	0	$277,852	$25,604	$0	$157,531	$14,517	$0	$541,002	$49,854	$0	$0
North & Central America	Mexico	6,456	23,581	3,018	$1,022,702	$3,735,260	$478,071	$600,831	$2,194,443	$280,864	$2,711,688	$9,904,020	$105,634	$120,724
North & Central America	Nicaragua	8,981	0	0	$1,577,173	$0	$0	$914,638	$0	$0	$3,772,188	$0	$0	$0
North & Central America	Panama	305	0	0	$50,850	$0	$0	$29,683	$0	$0	$128,268	$0	$0	$0
North & Central America	Puerto Rico	50	0	0	$11,468	$0	$0	$6,463	$0	$0	$21,000	$0	$0	$0
North & Central America	United States of America	55,326	15,889	455,172	$10,820,332	$3,107,438	$89,019,773	$6,197,926	$1,779,952	$50,990,852	$23,236,920	$6,673,296	$15,931,006	$18,206,864
South America	Argentina	0	0	12,183	$0	$0	$2,602,942	$0	$0	$1,477,191	$0	$0	$426,419	$487,336
South America	Bolivia (Plurinational State of)	802	359	0	$120,780	$54,017	$0	$71,437	$31,949	$0	$336,672	$150,570	$0	$0
South America	Brazil	0	0	47,600	$0	$0	$8,127,678	$0	$0	$4,729,535	$0	$0	$1,666,000	$1,904,000
South America	Chile	2,050	1,542	13,246	$224,588	$168,885	$1,450,992	$139,686	$105,040	$902,467	$861,084	$647,514	$463,600	$529,828
South America	Colombia	3,413	621	0	$566,680	$103,158	$0	$330,908	$60,238	$0	$1,433,460	$260,946	$0	$0
South America	Ecuador	1,546	51,958	0	$339,958	$11,424,507	$0	$192,377	$6,464,941	$0	$649,362	$21,822,234	$0	$0
South America	French Guiana	329	0	0	$47,478	$0	$0	$28,256	$0	$0	$138,348	$0	$0	$0
South America	Guyana	68	0	0	$10,213	$0	$0	$6,037	$0	$0	$28,350	$0	$0	$0
South America	Paraguay	0	0	35,491	$0	$0	$7,965,426	$0	$0	$4,498,506	$0	$0	$1,242,189	$1,419,645
South America	Peru	2,620	1,311	0	$428,189	$214,201	$0	$250,544	$125,334	$0	$1,100,526	$550,536	$0	$0
South America	Suriname	31	0	0	$4,819	$0	$0	$2,843	$0	$0	$13,188	$0	$0	$0
South America	Trinidad and Tobago	58	56	0	$9,922	$9,546	$0	$5,772	$5,553	$0	$24,360	$23,436	$0	$0
South America	Uruguay	352	0	9,400	$109,013	$0	$2,910,308	$59,929	$0	$1,599,934	$147,882	$0	$329,000	$376,000
South America	Venezuela (Bolivarian Republic of)	2,169	480	0	$369,983	$81,936	$0	$215,319	$47,684	$0	$910,896	$201,726	$0	$0
	Global totals	2,396,700	957,017	26,703,456	$408,327,232	$158,506,517	$4,503,965,422	$237,595,774	$92,566,401	$2,624,789,098	$1,006,614,070	$401,947,142	$934,620,957	$1,068,138,236